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Normal Force and Center of Pressure of a Clipped Delta Fin with Supersonic Leading Edges

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Introduction

Currently, LOW SUPERSONIC BARROWMAN EQUATIONS.xls models delta fin with tips clipped off and with subsonic leading edges. As in this code, the fin is assumed to be a flat plate with a clipped delta planform.

Reference 1 is the point of departure for this memo. Note that all the $\frac{\partial \Delta p}{\partial \alpha}$ used are averaged over their relevant regions.

Nomenclature

Mnemonic	Definition
b	Exposed semispan,
$C_{N\alpha}$	Fin panel normal force coefficient slope, rad^{-1},
c_R	Root chord,
c_T	Tip chord,
M	Free stream Mach number,
Δp	Pressure difference across the two sides of the fin,
q	Dynamic pressure,
S_{REF}	Aerodynamic reference area,
x_{CP}	Fin center of pressure, aft of the root chord leading edge,
α	Angle of attack,
Λ	Leading edge sweepback angle, and
μ	Mach angle.

Analysis

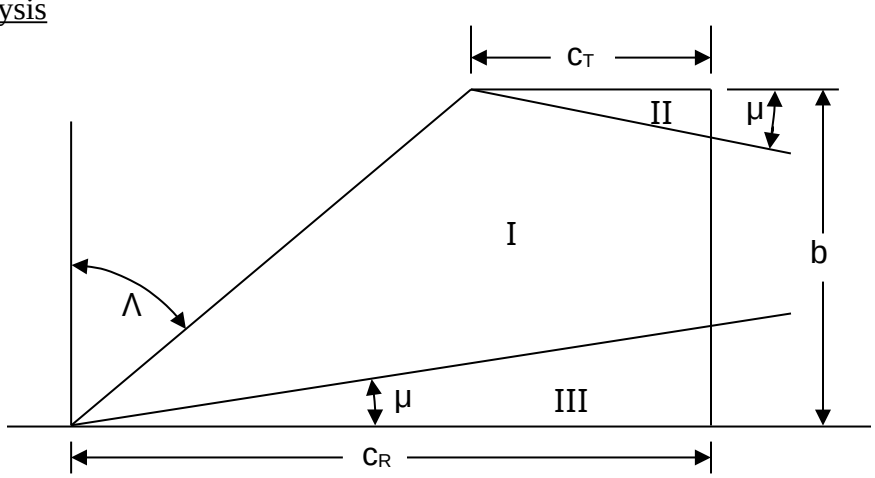


Figure 1 High Supersonic Mach number Fin Geometry

Figure 1 above shows a sketch of the relevant fin geometry The Mach angle is

$$Sin\mu = \frac{1}{M} \tag{1}$$

Normal Force Slope

Next, consider the average lifting pressure difference found in the three regions sketched above as $\frac{1}{q}\frac{\partial \Delta p}{\partial \alpha}$. The following equations are taken from Ref. (2). For Region I it is given by

$$\frac{1}{q}\frac{\partial \Delta p_I}{\partial \alpha} = \frac{4\cos\Lambda}{\sqrt{M^2\cos^2\Lambda - 1}}, \tag{2}$$

and for Region II

$$\frac{1}{q}\frac{\partial \Delta p_{II}}{\partial \alpha} = \frac{2\cos\Lambda}{\sqrt{M^2\cos^2\Lambda - 1}}, \text{and} \tag{3}$$

finally, for Region III:

$$\frac{1}{q}\frac{\partial \Delta p_{III}}{\partial \alpha} = \frac{4}{\sqrt{M^2\cos^2\Lambda - 1}}\left[\frac{\sqrt{M^2\cos^2\Lambda - 1}}{\tan\Lambda} - \cos\Lambda\left[\frac{\sqrt{M^2 - 1}}{\tan\Lambda} - 1\right]\right] \tag{4}$$

As the Mach number is reduced a small triangular region bounded by the Mach lines bounding Regions II and III forms on the trailing edge. Call this Region IV. According to Ref. (3)

$$\frac{1}{q}\frac{\partial \Delta p_{IV}}{\partial \alpha} = -\frac{1}{q}\frac{\partial \Delta p_I}{\partial \alpha} + \frac{1}{q}\frac{\partial \Delta p_{II}}{\partial \alpha} + \frac{1}{q}\frac{\partial \Delta p_{III}}{\partial \alpha}, \text{ or}$$

$$\frac{1}{q}\frac{\partial \Delta p_{IV}}{\partial \alpha} = \frac{4}{\sqrt{M^2\cos^2\Lambda - 1}}\left[\frac{\sqrt{M^2\cos^2\Lambda - 1}}{\tan\Lambda} - \cos\Lambda\left[\frac{\sqrt{M^2 - 1}}{\tan\Lambda} - 1\right]\right] - \frac{2\cos\Lambda}{\sqrt{M^2\cos^2\Lambda - 1}} \tag{5}$$

To estimate the fin normal force coefficient slope, it is necessary to estimate the areas of the various regions. The entire fin area is

$$S_T = \frac{1}{b}(c_R + c_T), \tag{6}$$

and the areas of regions II and III are

$$S_{II} = \frac{1}{2}c_T^2 \tan\mu, \tag{7}$$

$$S_{III} = \frac{1}{2} c_R^2 \tan\mu . \tag{8}$$

The condition for the existence of a Region IV is

$$b > (c_R + c_T)\tan\mu . \tag{9}$$

If there is no Region IV the Region I area is

$$S_I = \frac{1}{b}(c_R + c_T) - \frac{1}{2}\tan\mu(c_R^2 + c_T^2) \tag{10}$$

Then, the total fin normal force coefficient slope for no Region IV is

$$S_{REF} C_{N\alpha} = S_I \frac{1}{q}\frac{\partial \Delta p_I}{\partial \alpha} + S_{II}\frac{1}{q}\frac{\partial \Delta p_{II}}{\partial \alpha} + S_{III}\frac{1}{q}\frac{\partial \Delta p_{III}}{\partial \alpha} \tag{11}$$

For slightly lower Mach numbers, Region IV will exist. The formulas for normal force and area for Regions II and III remain unchanged. But, the area for Region I can be found after some grungy trig to be:

$$\text{and} \quad S_{IV} = \frac{\Delta b^2 \cos^2\mu}{2\sin 2\mu}, \text{ where} \tag{12}$$

$$\Delta b = b - (c_R + c_T)\tan\mu . \tag{13}$$

$$S_I = \frac{b^2}{2}\frac{\cos^2\Lambda\cos^2\mu - \sin^2\Lambda\sin^2\mu}{\cos^2\Lambda\sin 2\mu}, \tag{14a}$$

Alternatively, and simpler,

$$S_I = S_T - S_{II} - S_{III} + S_{IV} . \tag{14b}$$

The maximum Mach number for which there can be a Region IV is found to be

$$M_{MAX} = \sqrt{\left[\frac{c_R + c_T}{b}\right]^2 + 1} . \tag{15}$$

Then, the total fin normal force coefficient slope when the root & tip Mach lines cross is

$$S_{REF} C_{N\alpha} = S_I \frac{1}{q}\frac{\partial \Delta p_I}{\partial \alpha} + S_{II}\frac{1}{q}\frac{\partial \Delta p_{II}}{\partial \alpha} + S_{III}\frac{1}{q}\frac{\partial \Delta p_{III}}{\partial \alpha} - S_{IV}\frac{\partial \Delta p_I}{\partial \alpha} . \tag{16}$$

Finally, while the is no upper Mach number bound for this theory, there is a lower Mach number for which this theory is valid. It occurs when the Mach line emanating from the root chord leading edge intersects the tip chord trailing edge. This minimum Mach number is

$$M_{MIN} = \sqrt{\frac{c_R^2}{b^2} + 1}\,. \tag{17}$$

Results for a typical fin with an 8" root chord, a 3" tip chord and a 8" exposed semispan are shown in Fig. 2 below:

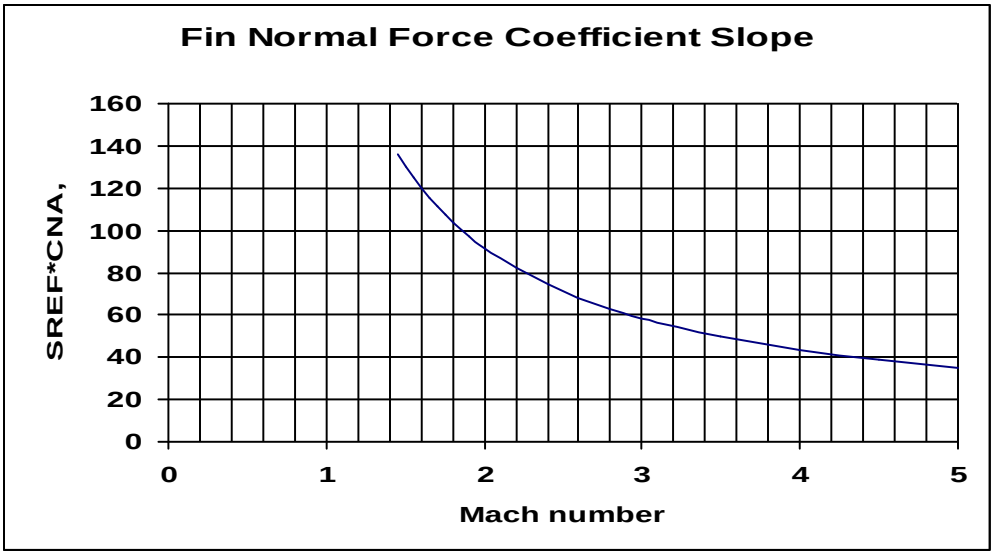


Figure 2. Typical Normal Force Slope

For the fin shown in Fig. 2, the theory is valid for Mach numbers greater than 1.4142, and there will be a Region IV for Mach numbers less than 1.7002.

Center of Pressure

The approach to estimating fin panel center of pressure is to find the Regional area pitch moments about the fin root chord leading edge. Begin by considering the high Mach number (no Region IV). First consider the entire fin without any root or tip losses:

$$Areamoment_T = \frac{b}{6}(2c_R^2 + 2c_R c_T - c_T^2) \tag{18}$$

Next, recall that the flow in the root and tip regions is conical, and therefore self-similar. The root region (III) has an

$$Areamoment_{III} = \frac{1}{3}c_R^3 \tan\mu\,, \text{ and} \tag{19}$$

for Region (II), $$Areamoment_{II} = \frac{1}{2}(c_R c_T^2 - \frac{1}{3}c_T^3)\tan\mu \,. \quad (20)$$

Then, the $$Areamoment_I = Areamoment_F - Areamoment_{II} - Areamoment_{III} \,. \quad (21)$$

Once the Mach number is reduced enough to create a Region IV, its

$$Areamoment_{IV} = (\frac{1}{3}c_R^3 \tan\mu)(c_R - \frac{\Delta b}{6\tan\mu}) \,. \quad (22)$$

The corresponding term for Region I is found per eq.(21):

$$Areamoment_I = Areamoment_T - Areamoment_{II} - Areamoment_{III} + Areamoment_{IV} \,. \quad (23)$$

The normal force moment about the root chord leading edge is:

$$NFmom = \sum Areamoment_I \frac{\partial \Delta p_I}{\partial \alpha} \,. \quad (24)$$

Finally, the fin x_{CP} is just

$$x_{CP} = \frac{NFmom}{S_{REF} C_{N\alpha}} \,. \quad (25)$$

Using the same typical fin numbers as before, the fin center of pressure is shown below in Fig.3.

Airfoil Effects

Finally, it is known that airfoil shape has a minor effect on $C_{N\alpha}$, but not on the center of pressure. The key is designing for a single wedge airfoil, one with a pointed leading edge and a blunt trailing edge. Reference (4) indicates about 10% enhancement to $C_{N\alpha}$. This leads to a multiplier applied to the above results.

References

1. W. C. House, "Aerobee-Hi (AJ11-6), "Fin Airloads", Aerojet Memo, 11 May 1954.
2. C. P. Hoult, "Aerodynamic Influence Coefficient Matrix for the Astrobee D Fins", Aerojet Memo 8500:M0629:ak, 9 April 1970.
3. R. P. Arnold, "Supersonic Aerodynamic Influence Coefficients for Fins", Space General memo:M0725:ak, 15 September 1972.
4. C. P. Hoult, "Supersonic Single Wedge Airfoil", RST memo, 2009.

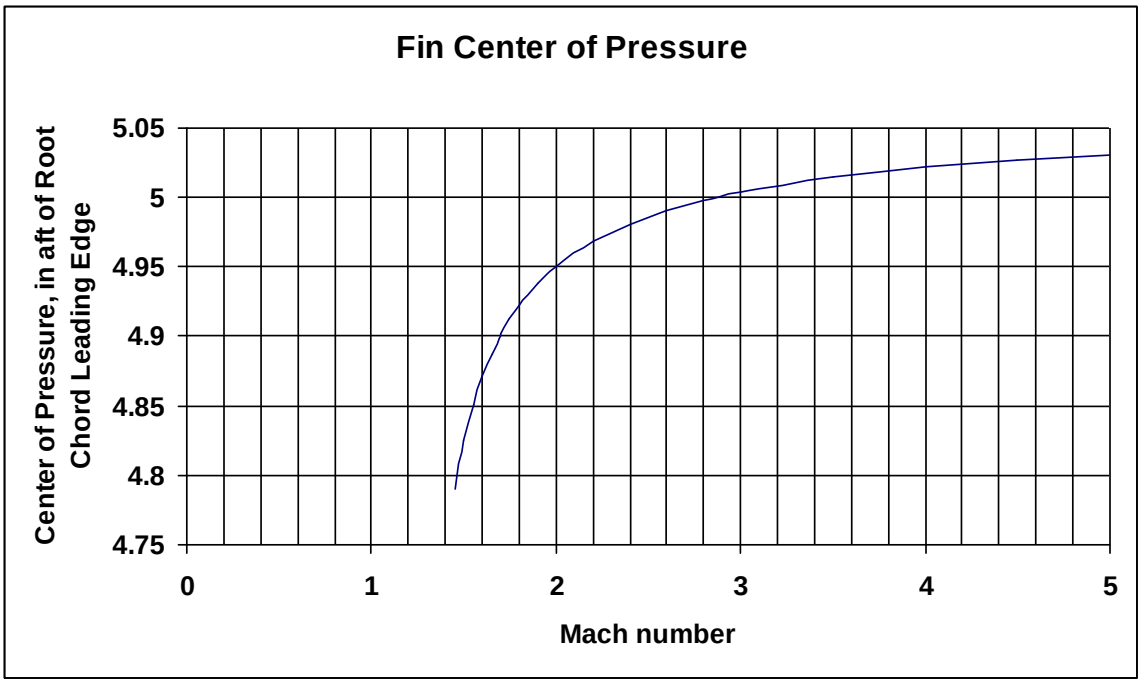


Figure 3 Typical Center of Pressure